OMER GADALLA

Aerospace Engineer

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EDUCATION

Middle East Technical University (METU) (2021-2026)

Bachelor of Aerospace Engineering

EXPERIENCE

Galaxy Rocket Team (Feb 2025-Present)

Propulsion Analysis Team Lead

TEKNOFEST Liquid Bipropellant Rocket Engine Development Competition (B3) — **Finalist 2025-2026**

- Collaborated with a 15-member team to design a liquid bipropellant engine achieving **100 N thrust**.
- Assessed in the design, analysis, and testing of a **10-bar** combustion chamber.
- Led **thermal, combustion, and CFD analysis**, validating combustion chamber design and performance at **3000K** .

CFD & Structural Analysis Team Lead

TEKNOFEST High Altitude Rocket Competition (A3) – 2025

- Contributed to the development of a **Mach 1.4, 15,000 ft** high-altitude rocket.
- Executed **CFD simulations** to calculate drag forces.
- Performed **FEA** structural Analysis **(static and dynamic)** to ensure integrity under high-altitude flight conditions.

MTS Defense Industry (July 2025-Sep 2025)

R&D Engineer – Intern

- Researched market needs in the Turkish space sector to identify satellite propulsion subsystem opportunities.
- Designed and proposed a **20N** green **bipropellant thruster** 50% faster than electric thrusters and 90% safer than hydrazine thrusters.

PROJECTS

CFD Analysis of a Converging-Diverging Nozzle (April 2025 – Jun 2025)

- Modeled nozzle flows under 7 pressure regimes in ANSYS Fluent and validated results against experimental data.

Combustion Analysis of a Hydrogen-Oxygen Rocket Combustion Chamber (April 2025 – May 2025)

- Implemented a MATLAB code for **combustion analysis** calculating the equilibrium composition of 6 species.

Rocket Nozzle Design (Dec 2024 – Jan 2025)

- Created a **MATLAB** code to design and optimize a rocket nozzle using **Method Of Characteristics** numerical technique.

CFD Analysis of Airfoils (NACA 0012 & 2412) (April 2024)

- Simulated lift/drag characteristics in **ANSYS Fluent** for aerodynamic performance evaluation.

TECHNICAL SKILLS

CAD: Inventor, FUSION 360, SOLIDWORKS || **Structural Analysis, Thermal Analysis**: ANSYS, Nastran

CFD: ANSYS Fluent || **Programming:** MATLAB, Simulink, Simscape, NASA CEA, Openrocket

Languages: Arabic (Native), Turkish (Fluent), English (Fluent)

CERTIFICATIONS

- MATHWORKS. Simscape Onramp
- Figes A.Ş. MATLAB & Simulink
- MATHWORKS. MATLAB Onramp
- MATHWORKS. Simulink Onramp